

THE ADAPTIVE COUPLING UPPER BOUND FOR THE WASSERSTEIN SEPARATION: MICROSCOPIC FLIPS AGAINST THE MACROSCOPIC SYMMETRY COST

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ABSTRACT. In the setting of Paper XXIX — extreme broken-phase KMS states $\varphi_{\pm}$ separated by the order parameter $\Psi = \chi(g) \prod_j \varepsilon_j(x_j)$, Lip-norm with weights w_0 (group direction) and w_j , geometric coordinates with parameters t_j — the only known upper bound on $W_1(\varphi_+, \varphi_-)$ was the trivial w_0 , paid by flipping the group coordinate everywhere. We construct an x -adaptive coupling and prove the closed-form nontrivial bound: for every enumeration (j_k) of J , with $\tau_j = \frac{2t_j}{1+t_j}$ and $m_j = \frac{1-t_j}{1+t_j}$,

$$W_1(\varphi_+, \varphi_-) \leq w_0 \Pi_{\infty}(\beta) + \sum_{k \geq 1} w_{j_k} \tau_{j_k} \prod_{l < k} m_{j_l},$$

the expected cost of the first available cheap flip in an independent scan; it is strictly below w_0 as soon as some $w_j < w_0$, is minimized by enumerating in increasing w_j (a three-line exchange lemma), and pinches against the squeeze lower bound $w_0 \Pi_{\infty}$ as $\beta \rightarrow \infty$, with explicit gap $O(\sum_j w_j t_j)$. The construction rests on an exact identity: the odd-conditioned geometric law is the unit shift of the even-conditioned one, so the parity channel at coordinate j transports at cost exactly w_j per unit mass, with capacity τ_j ; the residual mass m_j is matched exactly by the group flip and is, after renormalization, the extreme pair of the subsystem without coordinate j — whence the recursion. On the toy model of Paper XXIX's numerical experiment the bound evaluates to $5/12 \approx 0.4167$ against the certified window $[0.2254, 0.2288]$: consistent, decisively below $w_0 = 1$, and honestly a factor ≈ 1.8 from optimal near criticality, where the true plan shades amplitudes continuously rather than flipping the first available coordinate — the remaining gap is recorded.

1. SETTING, COST, AND THE TWO EXACT IDENTITIES

We keep the notation of Paper XXIX: $X = G \times \prod_{j \in J} \overline{\mathbb{N}}$, $\nu_{\text{sym}} = \text{Haar} \otimes \bigotimes_j \mu_{t_j}$ with $\mu_t(k) = (1-t)t^k$, $\varepsilon_j = (-1)^{x_j}$, $m_j = \mathbb{E}[\varepsilon_j] = \frac{1-t_j}{1+t_j}$, $\Psi = \chi(g) \prod_j \varepsilon_j$, $\nu_{\pm} = 2 \cdot \mathbf{1}_{\{\Psi = \pm 1\}} \nu_{\text{sym}}$, and the Lip-norm L with weights $(w_0, (w_j))$. By Kantorovich duality [2], $W_1(\varphi_+, \varphi_-) = \inf_{\gamma} \int d_L d\gamma$ over couplings $\gamma \in \Gamma(\nu_+, \nu_-)$, where $d_L(z, z') = \sup\{|f(z) - f(z')| : L(f) \leq 1\}$. We use only two evaluations, both immediate from the pointwise constraint: a group flip costs at most w_0 , $d_L((g, x), (g\sigma, x)) \leq w_0$, and a unit step in coordinate j costs at most w_j .

Lemma 1.1 (The shift identity). *Let $\mu = \mu_t$ and let μ^e, μ^o be its restrictions to even and odd integers, of masses $p_e = \frac{1}{1+t}$, $p_o = \frac{t}{1+t}$. Then the normalized laws satisfy, exactly,*

$$\frac{\mu^o}{p_o} = (\cdot + 1)_* \frac{\mu^e}{p_e} : \quad \frac{(1-t)t^{2k+1}}{p_o} = (1-t^2)t^{2k} = \frac{(1-t)t^{2k}}{p_e} \quad (k \geq 0).$$

Consequently the sub-measure $\theta := t \mu^e$ has mass p_o and $(\cdot + 1)_ \theta = \mu^o$: a parity-swapping transport inside one coordinate moves every unit of mass by exactly one step.*

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Paper XXXIII of the series on localized Bost–Connes systems. The framework is that of [1], cited by DOI; Paper XXIX of the series is referred to by numeral, and the setting is restated in place.

Proof. Both displayed computations are one line; $(\cdot + 1)_* \theta(2k + 1) = t\mu^e(2k) = (1 - t)t^{2k+1} = \mu^o(2k + 1)$. $\square$

Lemma 1.2 (Residual identity). *Fix $j \in J$ and condition on $(g, x_{\neq j})$. Write $s = \chi(g) \prod_{i \neq j} \varepsilon_i \in \{\pm 1\}$, so that ν_+ (resp. ν_-) puts x_j in parity s (resp. $-s$). Couple, on each fibre, the maximal common mass p_o by the shift of Lemma 1.1 (direction $+1$ on $s = +1$ fibres, -1 on $s = -1$ fibres; no boundary issue, odd values being ≥ 1). Then the residual measures are*

$$\nu_+^{\text{res}} = m_j \cdot \left[2 \cdot \mathbf{1}_{\{s=+1\}} (\text{Haar} \otimes \mu_{\neq j}) \right] \otimes \frac{\mu_{t_j}^e}{p_e}, \quad \nu_-^{\text{res}} = m_j \cdot \left[2 \cdot \mathbf{1}_{\{s=-1\}} \cdots \right] \otimes \frac{\mu_{t_j}^e}{p_e},$$

of total mass m_j each; and $m_j^{-1}(\nu_+^{\text{res}}, \nu_-^{\text{res}})$ is exactly the extreme pair of the subsystem on $G \times \overline{\mathbb{N}}^{J \setminus \{j\}}$ (order parameter s), tensored with a common spectator law in the j -th coordinate.

Proof. On a fibre with $s = +1$, ν_+ carries $x_j \sim \mu^e$ of mass p_e and the shifted sub-measure is $\theta = t\mu^e$, so the leftover is $(1 - t)\mu^e$, of mass $(1 - t)p_e = m_j$ and normalized law μ^e/p_e ; on $s = -1$ fibres ν_+ is exhausted and ν_- retains the same leftover. Reassembling over fibres gives the display; the identification with the subsystem pair is the equality $\Psi = s$ on the residual support ($\varepsilon_j \equiv +1$ there). $\square$

2. THE ADAPTIVE COUPLING AND THE BOUND

Theorem 2.1 (Adaptive coupling upper bound). *For every enumeration $(j_1, j_2, \dots)$ of J ,*

$$W_1(\varphi_+, \varphi_-) \leq w_0 \Pi_\infty(\beta) + \sum_{k \geq 1} w_{j_k} \tau_{j_k} \prod_{l < k} m_{j_l}, \quad \tau_j := 1 - m_j = \frac{2t_j}{1 + t_j},$$

the right side being the expected cost $\mathbb{E}[w_{J^*}]$ of an independent scan that at step k flips coordinate j_k with probability τ_{j_k} (cost w_{j_k}) and, if no coordinate ever flips (probability Π_∞), pays the group flip w_0 . The bound is strictly smaller than w_0 whenever some $w_j < w_0$, and

$$0 \leq \left(w_0 \Pi_\infty + \sum_k w_{j_k} \tau_{j_k} \prod_{l < k} m_{j_l} \right) - w_0 \Pi_\infty \leq 2 \sum_j w_j t_j \xrightarrow{\beta \rightarrow \infty} 0 :$$

the upper bound pinches against the squeeze lower bound $w_0 \Pi_\infty$ of Paper XXIX in the deep-freeze limit.

Proof. Apply Lemma 1.2 at j_1 : a coupling moving mass $1 - m_{j_1} = \tau_{j_1}$ by one j_1 -step (cost $\leq w_{j_1}$ each, Lemma 1.1) and leaving a residual pair of mass m_{j_1} which is m_{j_1} times the extreme pair of the subsystem without j_1 (the spectator coordinate is coupled by the identity, at zero cost). Iterate on $j_2, j_3, \dots$; after exhausting J the residual pair, of mass Π_∞ , is supported on $\{s' = +1\} \times$ vs $\{s' = -1\} \times$ with $s' = \chi(g)$, and the group flip $T(g, x) = (g\sigma, x)$ transports it exactly, at cost $\leq w_0$ per unit. Summing costs gives the display; each partial sum plus the residual masses telescopes to 1, whence the scan interpretation. Strictness: replacing the final w_0 -payment on the k -th channel's mass by the channel itself saves $(w_0 - w_{j_k})\tau_{j_k} \prod_{l < k} m_{j_l} > 0$. The gap estimate: $\tau_j \leq 2t_j$ and $\prod_{l < k} m_{j_l} \leq 1$. $\square$

Proposition 2.2 (Optimal enumeration). *The bound of Theorem 2.1 is minimized by enumerating J in increasing order of w_j .*

Proof. Swapping adjacent j, j' changes the bound by $w_j \tau_j + m_j w_{j'} \tau_{j'} - w_{j'} \tau_{j'} - m_{j'} w_j \tau_j = \tau_j \tau_{j'} (w_j - w_{j'})$, independently of the remainder; so any enumeration with an adjacent inversion is improved by sorting. $\square$

Remark 2.3 (Against the numerical verdict, and the remaining gap). On the toy model of Paper XXIX's experiment ($w_0 = 1$, $w_j = (j + 1)^{-1}$, $t_j = p_j^{-1}$, $N = 3$): sorted enumeration ($w = \frac{1}{4}, \frac{1}{3}, \frac{1}{2}$) gives $\frac{1}{12} + \frac{2}{3}[\frac{1}{6} + \frac{1}{2}(\frac{1}{12} + \frac{2}{3})] = \frac{5}{12} \approx 0.4167$, against the certified window $[0.2254, 0.2288]$ of the SOCP computation: consistent, and decisively below the trivial $w_0 = 1$. The physical reading is the one the construction embodies: near the transition the thermal activity $\tau_j = \frac{2t_j}{1+t_j}$ of each prime opens a microscopic channel — an arithmetic fluctuation of cost w_j — that substitutes for the macroscopic symmetry flip of cost w_0 ; in the deep freeze all channels close ($\tau_j \rightarrow 0$) and the group flip recovers its monopoly, which is why the bound pinches there. The factor ≈ 1.8 separating the bound from the numerical optimum near criticality is honest and structural: the scan plan flips the *first* available coordinate deterministically, while the SOCP optimizer shades amplitudes continuously across coordinates; closing the gap requires randomized multi-coordinate plans (fractional channel usage), and their optimization over the channel simplex is the remaining problem, now bracketed on both sides by closed forms.

REFERENCES

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